

Pediatric Traumatic Brain Injury

Section 1: Case Summary

Scenario Title:	Pediatric Traumatic Brain Injury
Keywords:	Traumatic brain injury, Peds, Trauma
Brief Description of Case:	An 18-month-old presents to the ED at a community hospital with a head injury following an unwitnessed fall from significant height with initial loss of consciousness. He presents irritable but stable, and subsequently develops signs of increased ICP progressing to herniation (secondary to an epidural hematoma)

Goals and Objectives	
Educational Goal:	Practice management of pediatric traumatic brain injury
Objectives: (Medical and CRM)	1. Lead team through care of critically ill infant including assigning roles, using closed-loop communication and using a shared mental model 2. Perform ATLS-based primary and secondary survey of a pediatric patient 3. Recognize signs of traumatic brain injury in a trauma patient 4. Manage TBI and increased ICP using evidence-based practices 5. Re-assess patient with physiologic parameters change 6. Consider non-accidental injury in pediatric trauma patients
EPAs Assessed:	TTD EPA #1 – recognizing the unstable/critically ill patient and subsequently mobilizing care from healthcare team to initiate basic life support interventions Foundations EPA #1 - initiating resuscitation of critically ill patients Core EPA #2 – resuscitating and coordinating care for a critically injured patient Core EPA #3 – providing airway management and ventilation

Learners, Setting and Personnel					
Target Learners:	☒ Junior Learners		☒ Senior Learners		☐ Staff
	☒ Physicians		☐ Nurses	☐ RTs	☐ Inter-professional
	☐ Other Learners:				
Location:	☒ Sim Lab		☐ In Situ		☐ Other:
Recommended Number of Facilitators:	Instructors: 1				
	Sim Actors: 1				
	Sim Techs: 1				

Scenario Development	
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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Sam		Age: 18 months		Gender: M	Weight: 12.5 kg
Presenting complaint: Head injury					
Temp: 37° C	HR: 145	BP: 90/50	RR: 35	O ₂ Sat: 98%	FiO ₂ : room air
Cap glucose: 4			GCS: 11 (E3 V3 M5)		
Triage note: Brought to ED by Mom after a fall off of playground in backyard (approximately 6 ft off ground). Mom heard a bang and then saw him on the ground. He had LOC x1 minute then woke up crying inconsolably. Mom did not see the fall as she was watching her other child inside the house. Vomited x1 en route to ED. Mom is currently out of the department giving her other child to grandparents in the parkade.					
Allergies: None					
Past Medical History: Previously healthy			Current Medications: None		

Section 2B: Extra Patient Information

A. Further History	
N/A	
B. Physical Exam	
Cardio: Tachycardic, capillary refill 2 seconds	Neuro: Pupils 4 mm and reactive. Quiet, moaning, cries with painful stimuli (i.e. IV insertion)
Resp: Clear breath sounds bilaterally	Head & Neck: Boggy hematoma left parietal scalp
Abdo: Normal	MSK/skin: Bruise on abdomen
Other: Normal	

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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (Toddler)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Gloves, PPE, stethoscope IV lines, IO set-up, IV push medications Nasal prongs, venturi mask, non-rebreather mask, bag valve mask, laryngoscope, ET tubes Cervical spine collar
C. Required Medications
Normal saline Mannitol, hypertonic saline Medications for RSI and sedation
D. Moulage
Boggy swelling to left parietal scalp
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed
☒ Patient not yet on monitor
F. Patient Reactions and Exam
Pupils 4 mm and reactive Crying with painful stimulus (i.e. IV insertion)

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Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
ED RN	Skillful and helpful. Reviews patient notes from triage, provides support with monitor application, IV start, medication administration, providing lab results. Key moments to prompt if not recognized by leader: recognition of head injury, deterioration in patient's GCS status, development of signs of increased ICP
Respiratory Therapist	Assists with airway support if requested



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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus tach HR: 145 BP: 90/50 RR: 35 O ₂ SAT: 98 % T: 37°C GCS: 11 AVPU: Verbal	Quiet, moaning, cries with painful stimulus (IV insertion)	<u>Expected Learner Actions</u> ☐ Place monitors ☐ Primary survey (ABCDE) including GCS and pupils ☐ Identifies skull injury and suspected TBI ☐ Immobilization of cervical spine ☐ Obtain 2 large bore PIVs ☐ Send trauma bloodwork ☐ Checks cap glucose	<u>Modifiers</u> - Labs to be provided if requested <u>Triggers</u> After 4 minutes or all actions complete → 2. Incr ICP (gradual progression toward state 2 vital signs)	As noted above, parent no longer present in ED therefore further patient information (beyond triage note) not available.
2. Increased ICP Rhythm: Sinus brady HR: 55 BP: 140/100 RR: 10 (irregular) O ₂ SAT: 93% GCS: 5 (E1 V1 M3) AVPU: Pain	Patient stops crying and GCS declines. Develops signs of Cushing's triad. Pupils: right 4 mm sluggish left 8 mm unreactive	<u>Expected Learner Actions</u> ☐ Re-examine patient given clinical change ☐ Identify signs of increased ICP ☐ Elevated head of bed ☐ Administer mannitol or hypertonic saline ☐ Assist respirations with BVM ☐ Prepare for intubation ☐ Consider seizure prophylaxis ☐ Call for clinical help (extra hands, neurosurgery, PICU) ☐ Ask for help finding mom	<u>Modifiers</u> - If no BVM initiated saturations drop to ~ 80% - If requests PICU back-up, not immediately available - If requests STAT CT, RN to prompt that patient needs to be stabilized <u>Triggers</u> Decision to intubate → 3. Intubation	No neurosurgical or ICU back-up available in this centre



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3. Intubation Rhythm: Sinus HR: 60 BP: 130/90 RR: 12 (irregular) O ₂ SAT: 93% GCS: 5 (E1 V1 M3)	Unchanged	<u>Expected Learner Actions</u> ☐ Verbalize airway plan including ventilation strategy ☐ Use Broselow tape to select appropriate size tube and depth ☐ Maintain inline stabilization ☐ Perform RSI (may consider neuroprotective adjuncts) ☐ Confirms tube placement	<u>Modifiers</u> - If mannitol or hypertonic saline initiated, left pupil begins to improve <u>Triggers</u> Intubation complete → 4. Disposition	
4. Disposition Rhythm: Sinus HR: 100 BP: 110/90 RR: per vent O ₂ SAT: 98% GCS: 2T AVPU: Unresponsive	Patient paralyzed. Improvement in vital signs. Pupils now 4 mm and reactive.	<u>Expected Learner Actions</u> ☐ Choose appropriate sedation ☐ Re-assess ABCs +/- secondary survey ☐ Order neuro and cervical spine imaging ☐ Verbalize targets for neuroprotective measures (T, BG, Na, pCO ₂) ☐ Arrange for transfer to tertiary care centre	<u>Modifiers</u> -Prompt to present the case for transport when PICU calls back (at the end of state 4) <u>Triggers</u> Arrange transport → END CASE	

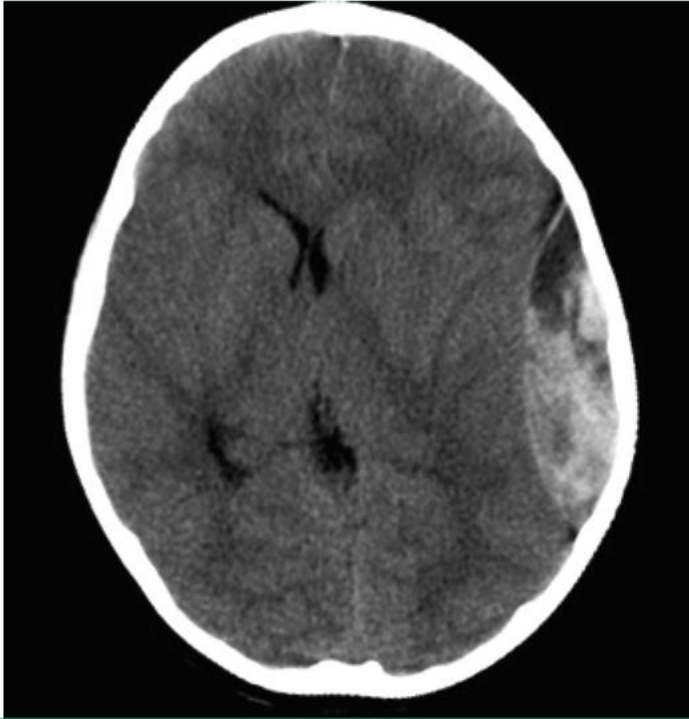
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Appendix A: Laboratory Results

<u>CBC</u> WBC 8 Hgb 130 Plt 200 <u>Lytes</u> Na 140 K 4.4 Cl 100 Urea 5 Cr 30 Glucose 4 <u>Extended Lytes</u> Ca 2.2 Mg 0.9 PO ₄ 1.5 <u>VBG</u> (start of case) pH 7.3 pCO ₂ 48 pO ₂ 80 HCO ₃ 22 Lactate 1	<u>Cardiac/Coags</u> INR 1 aPTT 30 <u>Biliary</u> AST 20 ALT 20 GGT 15 Bili 2 Lipase 10 <u>Tox</u> EtOH neg ASA neg Tylenol neg
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Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Epidural hematoma



From: Up-to-Date, Severe traumatic brain injury in children: Initial evaluation and management

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Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Questions for Discussion/Debrief

1. What are important signs of traumatic brain injury?
2. What are signs of impending cerebral herniation?
3. What is the current evidence re: hyperventilation in the setting of traumatic brain injury? (What are your targets?)
4. What are the targets for neuroprotective measures?
5. What RSI medications would you consider in cases of TBI and increased ICP?
6. How would you approach the concern of possible NAI given the report of unwitnessed injury?

References

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